

STRUCT MEMORY R1: Three Trainable Components for Long-term Memory Reasoning

Each component is trained via GRPO (Group Relative Policy Optimization) with outcome-based rewards

STAGE 2 — MEMORY WRITER

• TRAINABLE



Memory Manager
Learns structured CRUD operations over memory bank

▶ INPUT

- Current dialogue turn `d_t`
- Old memory bank `M_{t-1}`
- Extracted facts from turn `F_t`
- Structured memory tree (speakers / topics)

▶ OUTPUT

- Updated memory bank `M_t`
- JSON list of text, operations with `{id, text, event}`

⚙️ ACTION SPACE

ADD

UPDATE

DELETE

NOOP

CREATE_TOPIC

CREATE_NODE

MOVE

SPLIT

STAGE 3 — MEMORY READER

• TRAINABLE



Retrieval Agent
Learns to navigate structured memory tree for retrieval

▶ INPUT

- Question `q`
- Memory tree schema (speakers → topics → facts)
- Schema summary with entry counts

▶ OUTPUT

- Retrieval plan `{levels, selected_ids}`
- Retrieved memory entries for answer agent
- Early stopping signal

⚙️ ACTION SPACE

SELECT_SPEAKER

SELECT_TOPIC

SELECT_ENTITY

SELECT_IDS

STOP

STAGE 1 — MEMORY REASONER

• TRAINABLE



Answer Agent
Learns to generate answer from retrieved memory

▶ INPUT

- Question `q`
- Retrieved memories (grouped by speaker)
- Timestamps for temporal reasoning

▶ OUTPUT

- Memory distillation (relevant memory selection)
- Concise answer `a`

⚙️ ACTION SPACE

SELECT_MEMORY

REASON

GENERATE_ANSWER

file:///Users/liuyifan/Desktop/Research/Search-R1/figures/struct_memory_r1_figure.html

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